Solve mixed measure problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty? Copy or complete the first step.

2. A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?

3. Create two different calculations using 3/4, 0.6 and 25% that both give 30.

4. Miro says: Miro starts calculating before deciding what the problem asks and which information matters. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty? Copy or complete the first step.

   Answer 320 cm3.

2. A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?

   Answer 120 pupils.

3. Create two different calculations using 3/4, 0.6 and 25% that both give 30.

   Answer Answers vary and must preserve the stated values and give 30.

4. Miro says: Miro starts calculating before deciding what the problem asks and which information matters. Is this correct? Explain.

   Answer Represent the relationships first, label each subtotal and test the final answer against every condition.

Solve mixed measure problems

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ____________

1. A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?

2. A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?

3. Create two different calculations using 3/4, 0.6 and 25% that both give 30.

4. Solve this independently and show every stage: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?

5. Use a second method or representation for: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?

6. Check this result using an inverse, estimate or known fact: Create two different calculations using 3/4, 0.6 and 25% that both give 30.

7. Prove or explain your reasoning: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

8. Identify and correct this misconception: Miro starts calculating before deciding what the problem asks and which information matters.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Create two different calculations using 3/4, 0.6 and 25% that both give 30.

Teacher answers and checking prompts

1. A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?
   Answer 320 cm3.

2. A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?
   Answer 120 pupils.

3. Create two different calculations using 3/4, 0.6 and 25% that both give 30.
   Answer Answers vary and must preserve the stated values and give 30.

4. Solve this independently and show every stage: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?
   Answer 320 cm3.

5. Use a second method or representation for: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?
   Answer Expected result: 120 pupils. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Create two different calculations using 3/4, 0.6 and 25% that both give 30.
   Answer Expected result: Answers vary and must preserve the stated values and give 30. A valid check must be shown.

7. Prove or explain your reasoning: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.
   Answer 244.8. The explanation must show why the method works.

8. Identify and correct this misconception: Miro starts calculating before deciding what the problem asks and which information matters.
   Answer Represent the relationships first, label each subtotal and test the final answer against every condition.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Create two different calculations using 3/4, 0.6 and 25% that both give 30.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve mixed measure problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Prove or explain your reasoning: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

2. Prove the answer to this question, rather than only calculating it: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?

3. Solve this in two different ways and compare them: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?

4. Work backwards from the answer to reconstruct the question: Answers vary and must preserve the stated values and give 30.

5. Explain why this reasoning fails: Miro starts calculating before deciding what the problem asks and which information matters.

6. Create a new example that tests the same idea as: Create two different calculations using 3/4, 0.6 and 25% that both give 30.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.
   Answer 244.8. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?
   Answer Expected result: 320 cm3. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?
   Answer Expected result: 120 pupils. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: Answers vary and must preserve the stated values and give 30.
   Answer One valid reconstruction is: Create two different calculations using 3/4, 0.6 and 25% that both give 30. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro starts calculating before deciding what the problem asks and which information matters.
   Answer Represent the relationships first, label each subtotal and test the final answer against every condition.

6. Create a new example that tests the same idea as: Create two different calculations using 3/4, 0.6 and 25% that both give 30.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.